



SECR1213-12 (Network Communications)

LAB 2: Packet Tracer Simulation - Exploration of ARP and
Switch Table Communications

Name	Matric ID
Muhammad Afiq Danial bin Rozaidie	A23CS0117

Lecturer's Name: Sir Firoz Patel Dawoodi

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

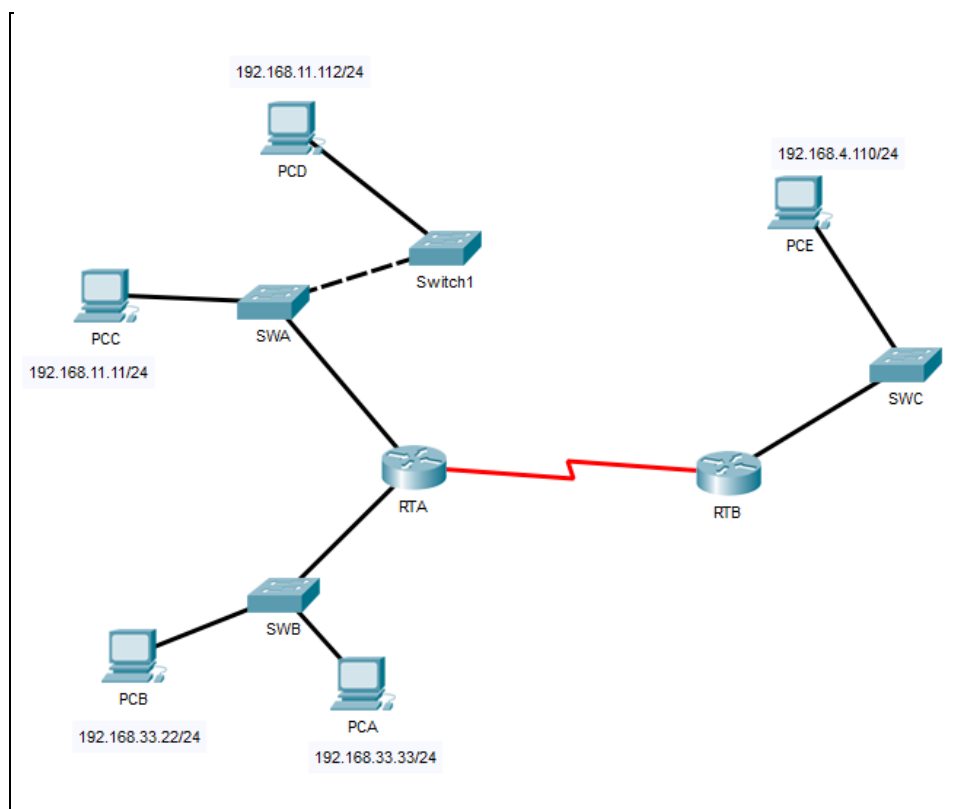


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.11.1      -         0002.4A00.0E91  ARPA   FastEthernet1/0
Internet 192.168.33.1      -         000C.CF0C.593A  ARPA   FastEthernet0/0
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.4.1      -         0001.977A.B614  ARPA   FastEthernet0/0
```

- c. At Switches SWA, SWB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

SWA

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0002.4a00.0e91    DYNAMIC     Fa0/1
1       000c.8546.7d85    DYNAMIC     Fa1/1
```

SWB

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a   DYNAMIC Fa0/1
```

SWC

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614   DYNAMIC Fa0/1
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type `arp -a` and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

PCA

```
C:\>arp -a
No ARP Entries Found
```

PCB

```
C:\>arp -a
No ARP Entries Found
```

PCC

```
C:\>arp -a
No ARP Entries Found
```

PCD

```
C:\>arp -a
No ARP Entries Found
```

PCE

```
C:\>arp -a
No ARP Entries Found
```

- e. What are your thoughts on the results?

No entry for all PCs because all the PCs doesn't receive any ARP request.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time=5ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms
```

- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746       dynamic
```

- In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06       dynamic
```

- In the Command prompt of PCC, PCD abd PCE, type **arp -a**. Paste the result of this command here.

PCC

```
C:\>arp -a
No ARP Entries Found
```

PCD

```
C:\>arp -a
No ARP Entries Found
```

PCE

```
C:\>arp -a
No ARP Entries Found
```

Step 2: Generate traffic between PCC to all other PC except PCA.

- a. Click **PCC** and click the Desktop tab > Command Prompt.
- b. Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
```

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126
Reply from 192.168.4.110: bytes=32 time=10ms TTL=126
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 4ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
192.168.11.112        0001.6462.0278        dynamic
```

- e. Discuss the results you got from all the commands on PCC.

The ARP entries are show in the ARP table only after PCC pings PCB, PCD, and PCE. No ARP entries are found before these ping activities. The ARP table displays entries once PCB, PCD, and PCE receive the ARP request from PCC and subsequently receive pings from PCB, PCD, and PCE.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11     3          00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22     3          0060.47EA.A746  ARPA   FastEthernet0/0
```

- g. At Router RTB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.4.1       -          0001.977A.B614  ARPA   FastEthernet0/0
Internet  192.168.4.110     2          0060.702D.7C08  ARPA   FastEthernet0/0
```


Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0001.6462.0278   DYNAMIC     Fa1/1
1       0002.4a00.0e91   DYNAMIC     Fa0/1
1       000c.8546.7d85   DYNAMIC     Fa1/1
1       00d0.d39a.c0d9   DYNAMIC     Fa2/1
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       000c.cf0c.593a   DYNAMIC     Fa0/1
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614   DYNAMIC Fa0/1
```

- d. Do switches use arp table? (Y/N)

Yes

- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge.*

Switches utilize an ARP table to store the IP addresses and corresponding MAC addresses of network devices. This table serves the purpose of identifying the destination MAC addresses for communication with network nodes.

- f. What information does the command `show mac-address-table` gives?

The command `show mac-address-table` displays the MAC address table of a switch, which includes information such as learned MAC addresses, associated VLANs, corresponding switch ports, and the type of entry (dynamic or static).

Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- Change the Lab 4 filename to include your name. *Example: Lab4AliAhmad.pkt
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot shows the Cisco Packet Tracer interface. On the left, a network topology is displayed with a switch (S1) connected to a wireless router (WRS2) and three PCs (PC1, PC2, PC3). The switch has three subinterfaces: G0/0.10 (172.17.10.1/24), G0/0.20 (172.17.20.1/24), and G0/0.88 (172.17.88.1/24). PC1 is connected to S1 F0/11 (172.17.10.21/24, VLAN 10). PC2 is connected to S1 F0/18 (172.17.20.22/24, VLAN 20). PC3 is connected to WRS2 (172.17.40.100/24). The PT Activity window on the right is titled "Packet Tracer – Configuring Wireless LAN Access" and shows an "Addressing Table" with the following data:

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

The PT Activity window also shows "Time Elapsed: 00:12:52" and "Completion: 100/100".

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

The screenshot shows the Cisco Packet Tracer interface. On the left, a network topology is displayed with a switch (S1) connected to a wireless router (WRS2) and three PCs (PC1, PC2, PC3). The switch has three subinterfaces: G0/0.10 (172.17.10.1/24), G0/0.20 (172.17.20.1/24), and G0/0.88 (172.17.88.1/24). PC1 is connected to S1 F0/11 (172.17.10.21/24, VLAN 10). PC2 is connected to S1 F0/18 (172.17.20.22/24, VLAN 20). PC3 is connected to WRS2 (172.17.40.100/24). The PT Activity window on the right is titled "Packet Tracer – Configuring Wireless LAN Access" and shows an "Addressing Table" with the following data:

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1
PC2	NIC	172.17.20.22	255.255.255.0	172.17.20.1
PC3	NIC	DHCP Assigned	DHCP Assigned	DHCP Assigned
WRS2	NIC	172.17.88.25	255.255.255.0	172.17.88.1

The PT Activity window also shows "Time Elapsed: 00:33:00" and "Completion: 100/100".